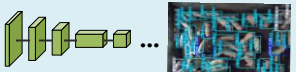





Video Stream



Edge Detection Model M_e^t

Model Degradation

Behavior-Aware Sample Selection

Selected Samples D_u^t

Behavior-aware Sample Selection

M_e^t

Initiazation Model Selection

Initialzation Model $M_h^{k^*}$

Historical Model Bank

Deploy Model

Store Accepted Model

Initialzation Model M_{init}^t

Source Replay D_{src}

Teacher Model

Pseudo_label $\hat{y}_i$

Model Training

Candidate Model M_{cand}^{t+1}

D_u^t

Cloud-side Continual Learning

M_{cand}^{t+1}

Shadow Validation (on holdout frames D_{val})

Adaptive Deployment

Validation and Deployment